

TEST CASE DOCUMENT

Springfield Public Library

Sub-Project: Member Registration & Authentication System

1. Test Overview

The purpose of this testing activity is to validate the functionality of the Online Member Registration and Secure Login features.

Tests ensure:

- Users can register successfully
 - Errors are handled properly
 - Form validation works as expected
 - Data is stored correctly in Firebase Authentication & Firestore
-

2. Test Environment

Component	Version / Info
Frontend	React.js
Backend	Firebase Authentication, Next.js
Browsers	Opera
OS	Windows 11
Test Data Source	Manually created user accounts

3. Test Scenarios & Test Cases

A. REGISTRATION FEATURE

Test Case 1 – Successful Registration

Field	Details
Title	Successful user registration
Pre-condition	User does not already exist in Firebase Auth
Steps	1. Open registration page. 2. Fill valid name, email, phone number, library pin. 3. Click "Submit". Spring Field Library Registration Form John Busko johnbusko96@yahoo.com 5558136475 2005 - 08 - 08 0000 Submit Login
Expected Result	- Firebase creates new account records. - UI displays: "Registered Successfull! Welcome..." and the library card number used to login in the future."

Actual Result

**Spring Field Library
Registration Form**

Registered Successfull!

welcome, John Busko
Your Library Card Number is 72779728

72779728

Please use the library card to log in.

[Login](#)

members	>	giECTxrnc9sVjkmSYgUM	+ Add field
		q2eeioFGYk6d82yFIK11	
			dob: "2005-08-08"
			email: "johnbusko96@yahoo.com"
			fullName: "John Busko"
			libraryCardNumber: "72779728"
			phone: "5558136475"
			pin: "0000"

Status PASS

Defects Found None

Test Case 2 – Missing Required Fields

| Steps | Submit form without filled in fields |
| Expected Result | “All fields are required” or React/Firebase form validation error |

**Spring Field Library
Registration Form**

All fields are required

Full Name

Email Address

Phone Number (10 digits)

yyyy - mm - dd

PIN (4 digits)

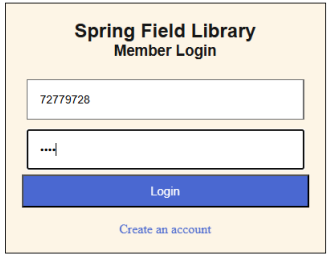
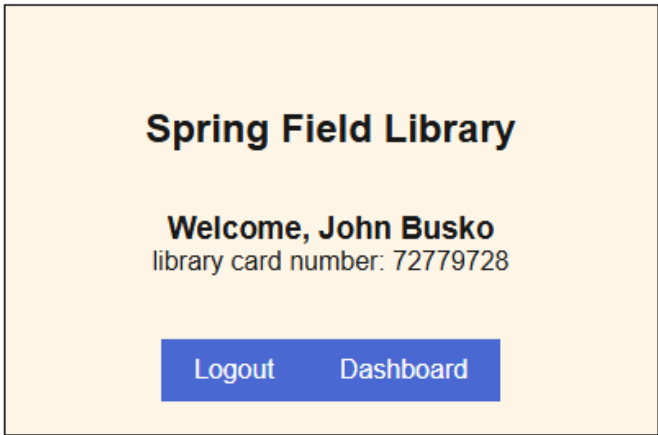
Submit

[Login](#)

| Actual Result |

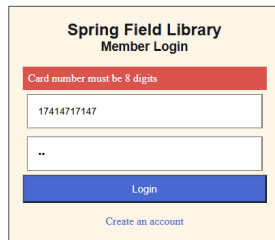
B. LOGIN FEATURE

Test Case 1 – Successful Login (Approved User)

Field	Details
Pre-condition	User is approved (status = “approved”)
Steps	1. Navigate to login. 2. Enter valid Library card number+PIN. 3. Submit. 
Expected Result	Displays user is successfully logged in and is prompted to either Member Dashboard or Logout.
Actual Result	
Status	PASS

Test Case 2 – Wrong Library Card Number

- | Steps | Enter correct PIN and wrong library card number |
- | Expected Result | UI shows: “Card number must be 8 digits.” |

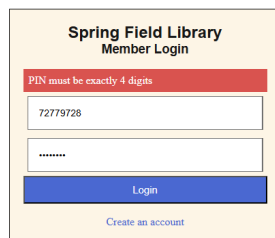


The screenshot shows the 'Spring Field Library Member Login' form. A red error message at the top states 'Card number must be 8 digits'. The 'Library Card Number' field contains the text '17414717147'. The 'PIN' field contains two asterisks. The 'Login' button is blue, and the 'Create an account' link is in blue text at the bottom.

- | Actual Result |
 - | Status | PASS |
-

Test Case 3 – Wrong PIN

- | Steps | Enter correct library card number and wrong PIN |
- | Expected Result | Display returns “PIN must be exactly 4 digits.” |

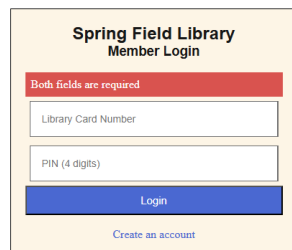


The screenshot shows the 'Spring Field Library Member Login' form. A red error message at the top states 'PIN must be exactly 4 digits'. The 'Library Card Number' field contains the text '72779728'. The 'PIN' field contains seven asterisks. The 'Login' button is blue, and the 'Create an account' link is in blue text at the bottom.

- | Actual Result |
 - | Status | PASS |
-

Test Case 4 – Empty Fields

- | Steps | Leave blank |
- | Expected Result | Validation error + Display “Both fields are required” message |



The screenshot shows the 'Spring Field Library Member Login' form. A red error message at the top states 'Both fields are required'. The 'Library Card Number' and 'PIN (4 digits)' fields are empty. The 'Login' button is blue, and the 'Create an account' link is in blue text at the bottom.

- | Actual Result |
 - | Status | PASS |
-

4. Defect Summary

Defect ID	Test Case	Description	Severity	Status
D1	B2	Error not displaying when number entered is in a valid format but is unregistered.	Medium	Fixed
D2	B3	The site crashes when the correct PIN format is entered but does not match the registered user's PIN.	High	Fixed

5. Conclusion

This test cycle validates the core functional requirements for:

- Account creation
- Authentication
- Pending → approved workflow
- Error handling

All defects discovered will have been fixed in the final submission, and this document supports ongoing quality monitoring throughout development.